<u>SBC Comparisons</u>

Board	Pros	Cons
RDK X5 	- Powerful AI acceleration (10TOPS) - Wi-Fi 6 + BT 5.4 - Rich multimedia interfaces - Compact and efficient	- Less community support than Raspberry Pi - Limited documentation **for beginners**
Raspberry Pi 5 	- Huge community and support - Dual 4K display support - PCIe expansion - Affordable and versatile	- Needs active cooling - Limited AI acceleration compared to Jetson
Jetson Nano 	- Excellent for AI and deep learning - CUDA support for TensorFlow, PyTorch - Low power consumption	- No onboard Wi-Fi - Limited general computing power - Fixed 4GB RAM
LattePanda Delta 3 	- Runs full Windows/Linux OS - Arduino-compatible GPIO - Powerful x86 CPU - NVMe and SATA support	- Higher power draw - Larger form factor - GPIO labelling can be confusing

Feature	RDK X5	Raspberry Pi 5	Jetson Nano	LattePanda Delta 3
CPU	Octa-Core Cortex-A55	Quad-core Cortex-A76 (2.4GHz)	Quad-Core Cortex-A57 (1.43GHz)	Intel Celeron N5105 (2.0-2.9GHz)
GPU	32GFlops GPU 10Tops BPU	Broadcom VideoCore VII	128 code NVIDIA Maxwell GPU	Intel UHD Graphics
RAM	4GB/8GB LPDDR4	4GB/8GB LPDDR4	4GB LPDDR4	8GB LPDDR4
Storage	1GB NAND + TF card slot	microSD + PCIe	microSD+ USB	64GB eMMC + NVMe/SATA Support
Wireless	Wi-Fi 6 + BT 5.4	Wi-Fi + BT	No onboard Wi-Fi	Wi-Fi 6 + BT 5.2
USB ports	4x USB 3.0	2x USB 3.0 2x USB 2.0	4x USB 3.0	3x USB 3.2 USB C
Display output	HDMI MIPI DSI	Dual 4K HDMI	HDMI DP via GPIO	HDMI + DP + eDP
Camera Support	2 x MIPI CSI	Dual CSI	1 x MIPI CSI-2	Arduino Compatible GPIO
Power input	5V/5A USB-C	5V/5A USB-C	5V/4A via micro-USB	12V DC or USB-C
OS support	Ubuntu 22.04 RDK OS Linux	Raspberry Pi OS Linux	Ubuntu-Based Jetpack	Windows 10/11 Linux

Board	GPIO count and Features
RDK X5	28 GPIOs, supports SPI, I2C, I2S, PWM, UART
Raspberry Pi 5	40 GPIOs, full support for SPI, I2C, UART, PWM, GPIO interrupts
Jetson Nano	40 GPIOs, supports I2C, UART, SPI, PWM, I2S
LattePanda Delta 3	Up to 100 GPIOs via ATmega32U4 (Arduino Leonardo), supports analog/digital I/O

Board	Dimensions (mm)	mm^3	Notes
RDK X5	85 × 56 × 20	95.2	Compact and stackable, ideal for embedded robotics and small form factor
Raspberry Pi 5	85 × 56 × 17	81	Same footprint as Pi 4; wide accessory compatibility
Jetson Nano	100 × 80 × 29	232	Larger due to heatsink and carrier board
LattePanda Delta 3	125 × 78 × 16	156	Slim but wide, includes Intel CPU and Arduino coprocessor

Best choice	Use case
RDK X5	Edge AI + Robotics
Raspberry Pi 5	General Maker Projects
Jetson Nano	AI/ML Projects
LattePanda Delta 3	Windows + Arduino Combo